Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech | *Time: 03 Hours*
Code: 105602 | *Subject: Machine Learning* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option for the following questions (Answer any seven):*** **[2 x 7 = 14]**

(a) Gradient Descent is an example of
(i) Evolutionary algorithm (iii) Optimization algorithm
(ii) Sampling algorithm (iv) Clustering algorithm

(b) Which regression technique performs feature selection?
(i) Ridge Regression (iii) LASSO
(ii) Linear Regression (iv) Polynomial Regression

(c) PCA chooses components that
(i) Maximize variance (iii) Minimize variance
(ii) Maximize covariance (iv) Increase dimensionality

(d) In Bayesian Learning, a conjugate prior is chosen because it
(i) simplifies posterior calculation (iii) improves accuracy
(ii) eliminates noise (iv) selects features

(e) Logistic Regression uses
(i) Linear activation (iii) Sigmoid activation
(ii) ReLU activation (iv) hyperbolic tangen

(f) Perceptron works only on
(i) Linear separable data (iii) Non-linear data
(ii) High-dimensional data (iv) Unsupervised data

(g) Bagging mainly helps in reducing
(i) Bias (iii) Noise
(ii) Variance (iv) Over-sampling

(h) K-medoids is more robust to
(i) Large data (iii) Outliers
(ii) Gaussian distribution (iv) High variance

(i) Bayesian networks represent
(i) Undirected dependencies (iii) Directed acyclic graph
(ii) Fully connected graph (iv) Weighted graph

(j) K-means uses
(i) Manhattan distance (iii) Centroids
(ii) Medoids (iv) Probabilities

Q.2 (a) Define hypothesis space and inductive bias with examples. **[7]**
(b) Explain the difference between supervised, unsupervised, and reinforcement learning with real world applications. **[7]**

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Q.3	(a)	What is bias–variance trade-off? Describe K-fold cross-validation and its importance.	[7]
	(b)	Draw the perceptron network with the notation. Derive an equation of gradient descent rule to minimize the error.	[7]
Q.4	(a)	Explain PCA with derivation of eigen decomposition and dimensionality reduction.	[7]
	(b)	Compare ML, MAP and Bayes estimators using an example.	[7]
Q.5	(a)	Describe ANN architecture and derive back propagation equations.	[7]
	(b)	Explain Decision Tree learning using Gini Index & Information Gain.	[7]
Q.6	(a)	Explain ensemble learning. How do Bagging and Boosting reduce error?	[7]
	(b)	Discuss hypothesis-testing and its role in ML model comparison.	[7]
Q.7	(a)	Compare K-means, K-medoids, and spectral clustering in detail.	[7]
	(b)	Derive the Expectation–Maximization algorithm step-by-step.	[7]
Q.8	(a)	Explain reinforcement learning with policy, reward, and value functions.	[7]
	(b)	Describe Bayesian Networks and inference techniques.	[7]

Q.9 Write Short notes on *any two* of the following: [2 x 7 = 14]

(a) Support Vector Machine
(b) linear regression
(c) Gaussian Mixture Model
(d) AdaBoost algorithm

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